

In-Car Conversational AI

Hands-free cabin interaction for a smarter driving experience

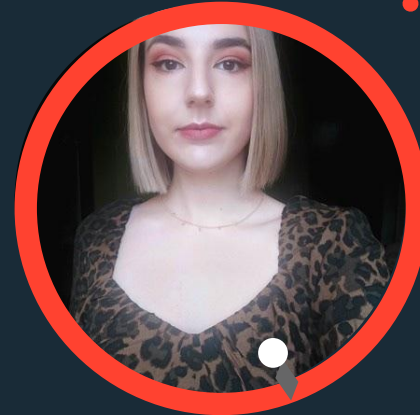
Voice AI • Face Recognition • Cabin Control • Navigation • Music

Team: Backseat Boys



Meet the Team

Gabriela Neagu
Mentor



Ioan Agape
Buddy



Nicolae Taradaciuc
Member



Andrei Murgulet
Member



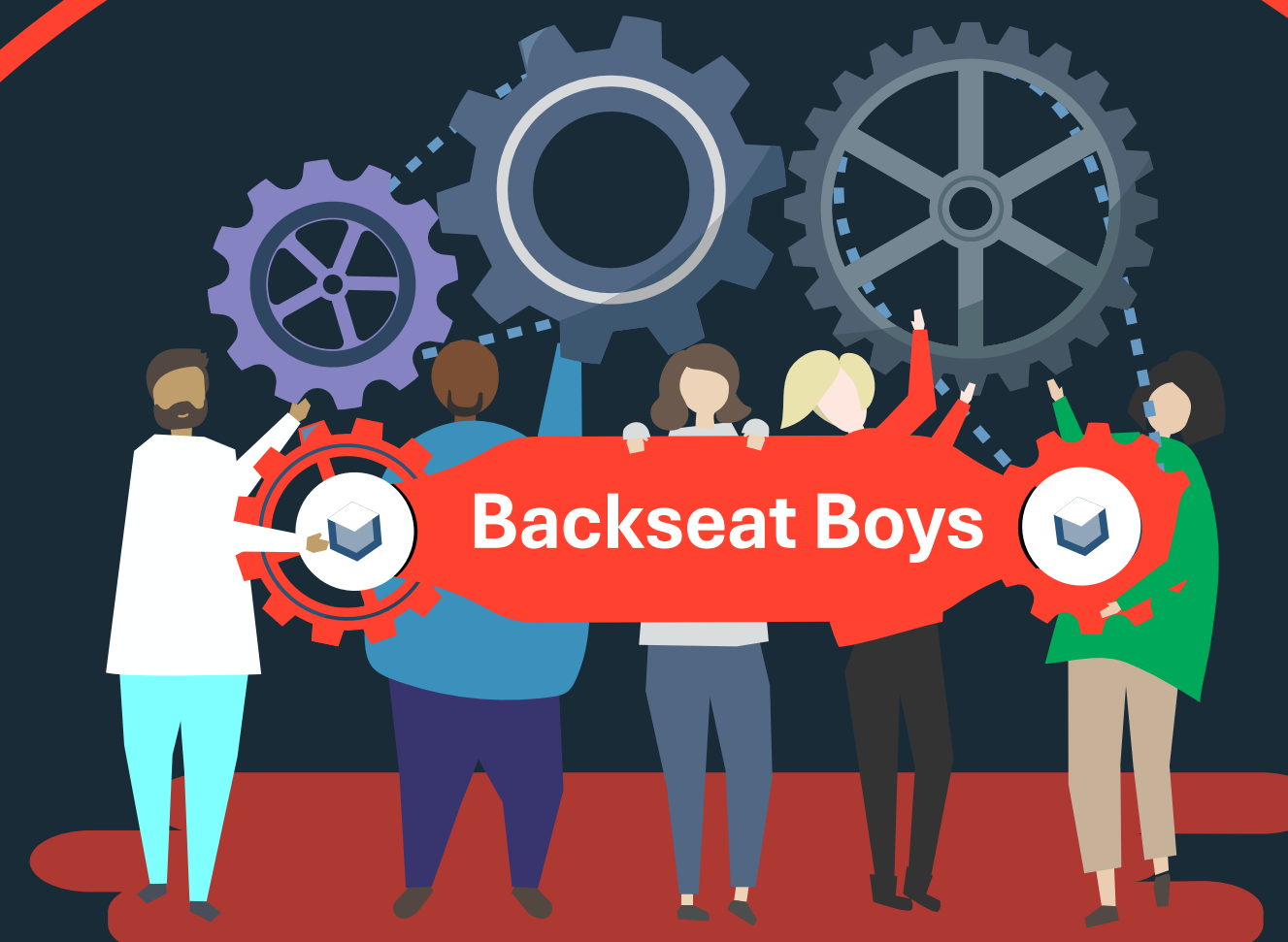
Mihai Ginghina
Member



Eusebiu Vatamaniuc
Member



Casian Bahrim
Member



In-Car Conversational AI

Project Brief

An in-car conversational AI that enables drivers and passengers to control non-driving features such as climate, fan speed, seats, lighting, and infotainment, through hands-free natural-language commands. The goal is to execute these commands reliably while minimizing driver distraction and improving the overall user experience.

How the command flows from speech to action

1

Voice Command

“Set the temperature to 22°C”

2

Conversational AI

Understands intent and user preference

3

Cabin / Infotainment Action

Climate · Fan · Seats · Lighting · Music · Navigation

From
Problems:

Driver Distraction

Manual adjustments can take attention away from the road during driving and reduce safety.



Slow Task Execution

Simple comfort or infotainment changes often require multiple taps, menus, and extra driver effort.



Unnatural Experience

Traditional controls feel less intuitive because users must adapt to menus instead of speaking naturally.



**From Problems
To Solutions**

To
Solutions:

Hands-Free Interaction

Users can control cabin and infotainment features through natural voice commands without touching the dashboard.



Faster Access

Common actions become quicker by letting users simply say what they want to change.



Natural Conversation

The system makes car interaction easier, more intuitive, and closer to a natural conversation.



Securely identifies the user when entering the car and loads the personalized system experience.

Face Recognition Login

Hey Rhaaspy

Listens for the wake word and starts voice command capture without manual interaction.

Uses AI to transcribe speech, understand user intent, and generate actionable system responses.

AI Voice Understanding

Controls temperature, fan speed, seat heating, and comfort settings through one interface.

Cabin Climate & Seat Comfort

Main Application Features

Supports music search and playback, extending the assistant into infotainment control.

Music Module

Provides maps, destination search, route visualization, and driving-oriented navigation views.

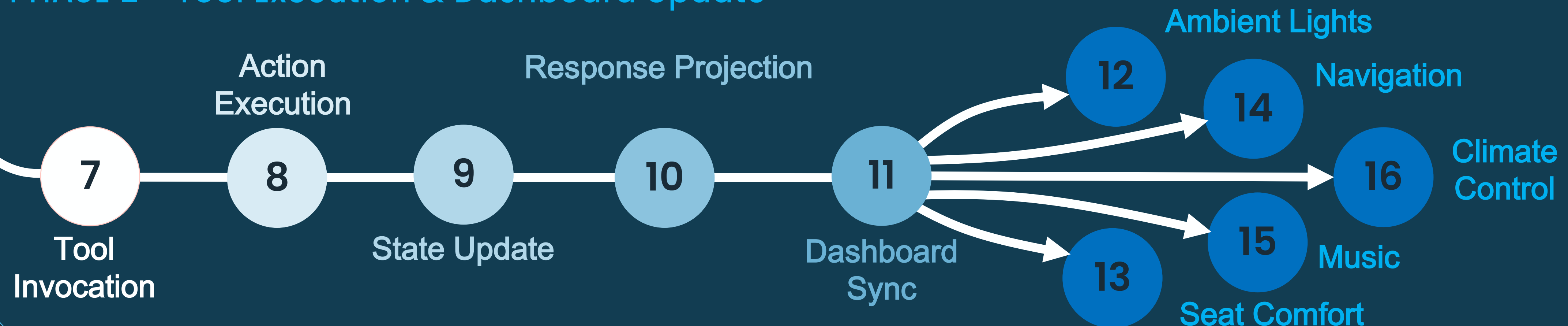
Navigation Module

PHASE 1 – AI Pipeline



6 Intent-to-Tool Routing

PHASE 2 – Tool Execution & Dashboard Update

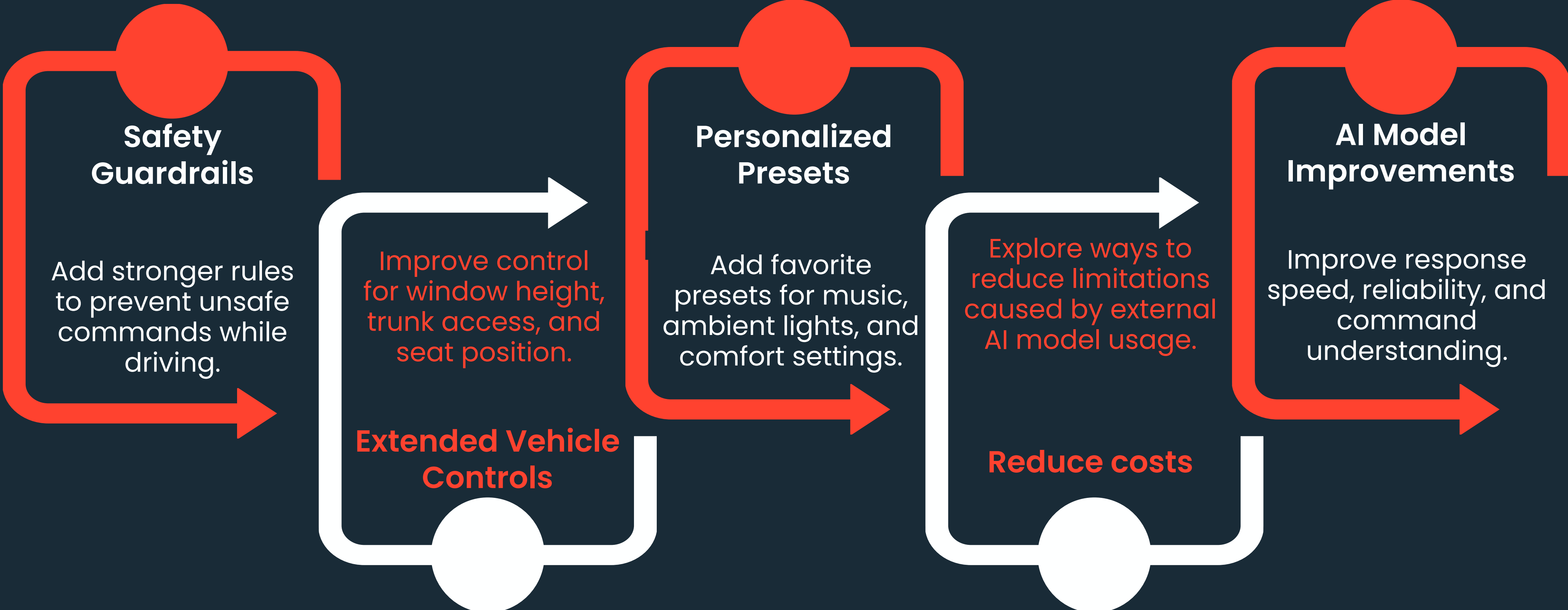


LIVE DEMO



Backseat Boys

What's next ?



Questions



Conclusions

Thank *you*!

Backseat
Boys